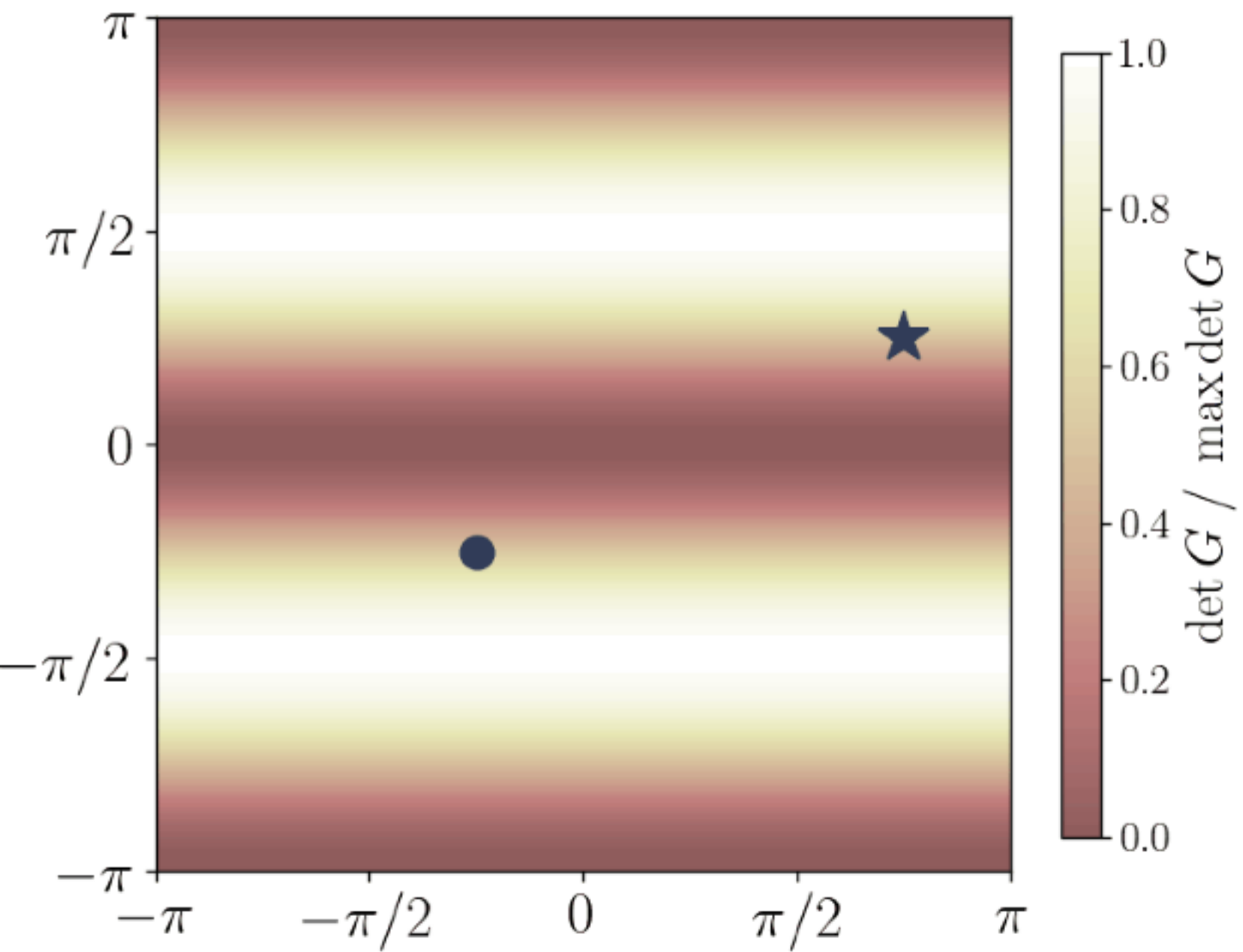






TWO-LINK PLANAR ARM

The simplest yet elegant anisotropic case

— **minimum-energy motions, kinetic-energy Riemannian metric**





Configuration space

TaskSpace

Euclidean vs. kinetic-energy

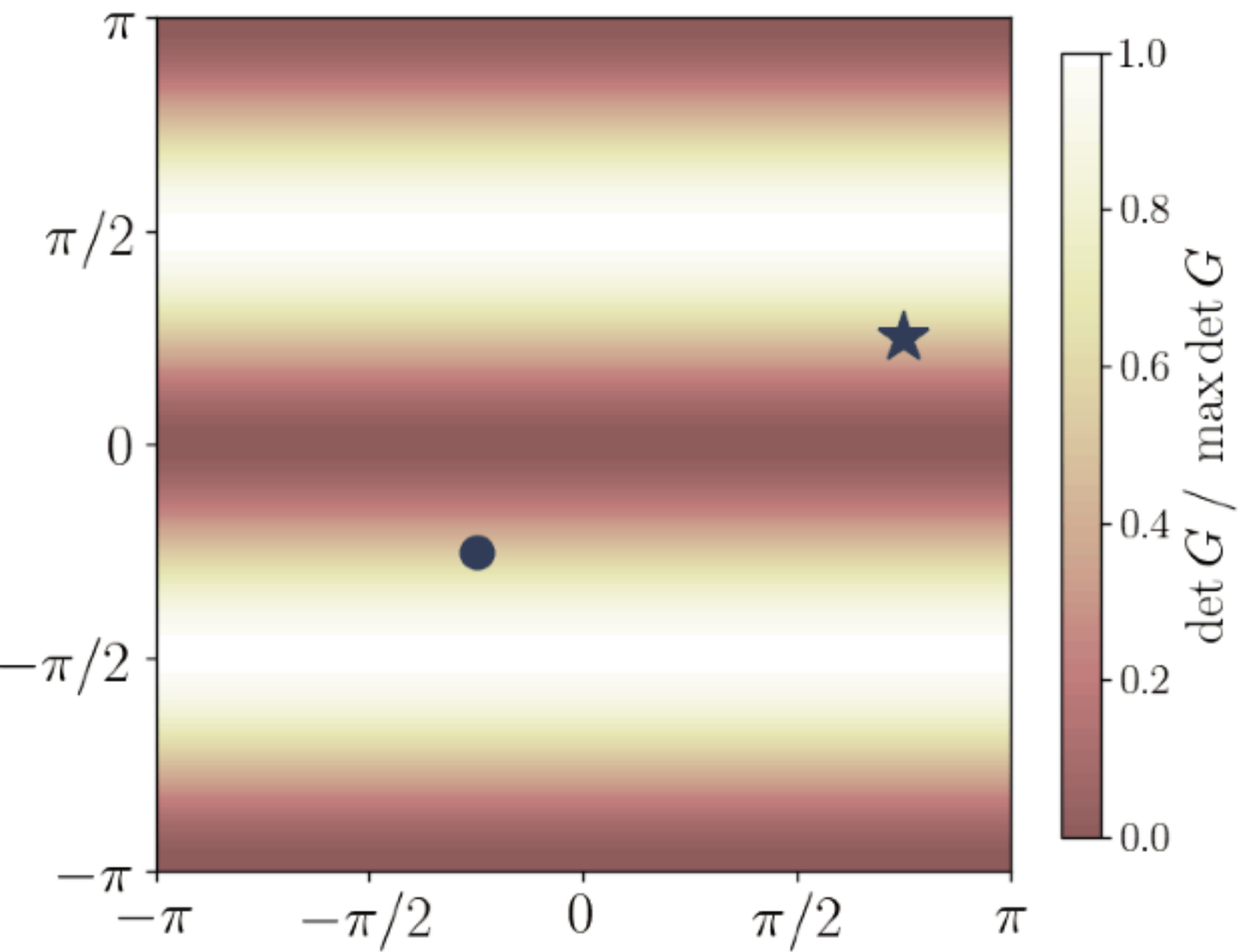
$RR T^*$ under the kinetic-energy metric

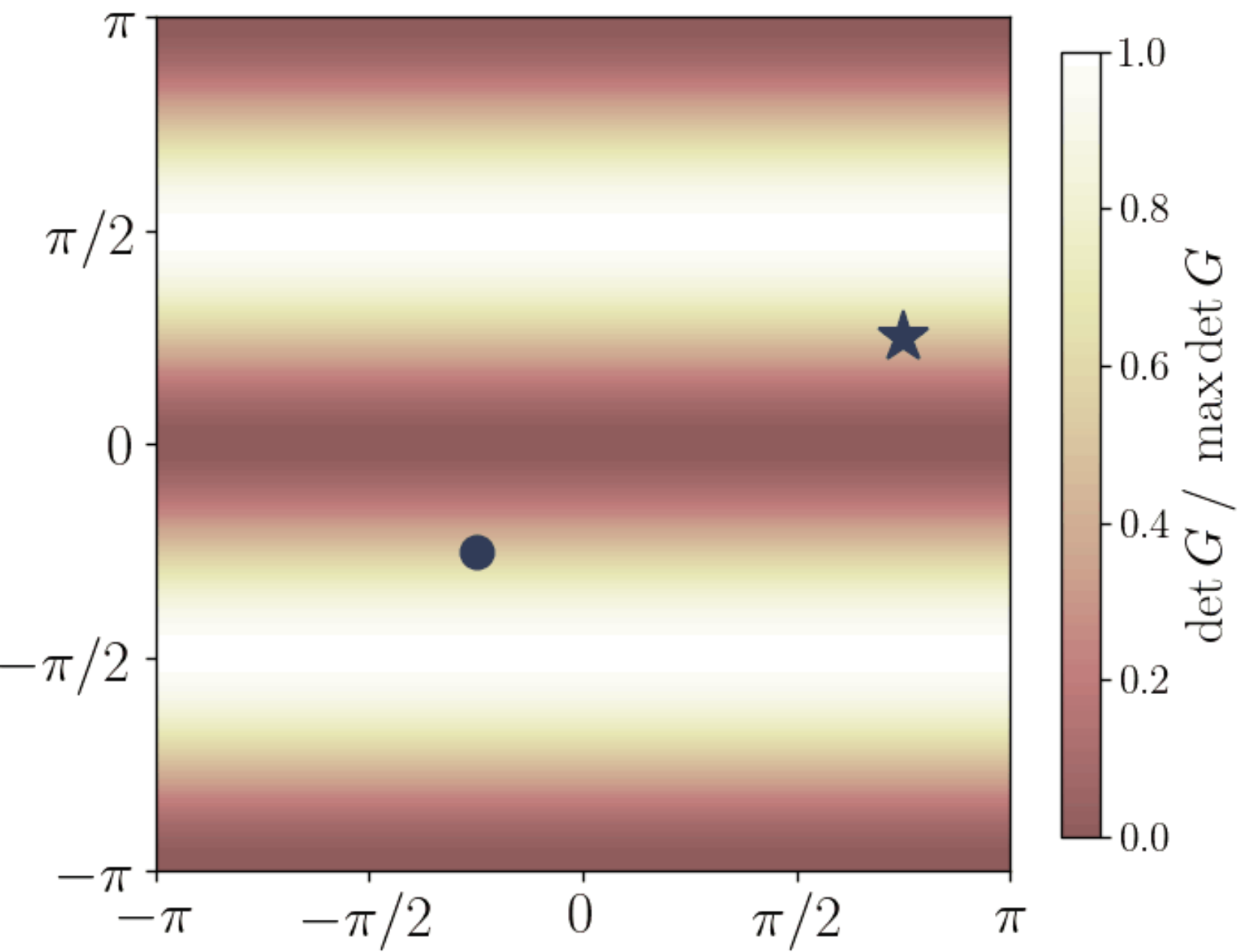
$$T(q,\dot{q})\equiv \frac{1}{2}\dot{q}^\top M(q)\dot{q}\equiv \Rightarrow G(q)\equiv M(q)$$


inertia matrix

1

5







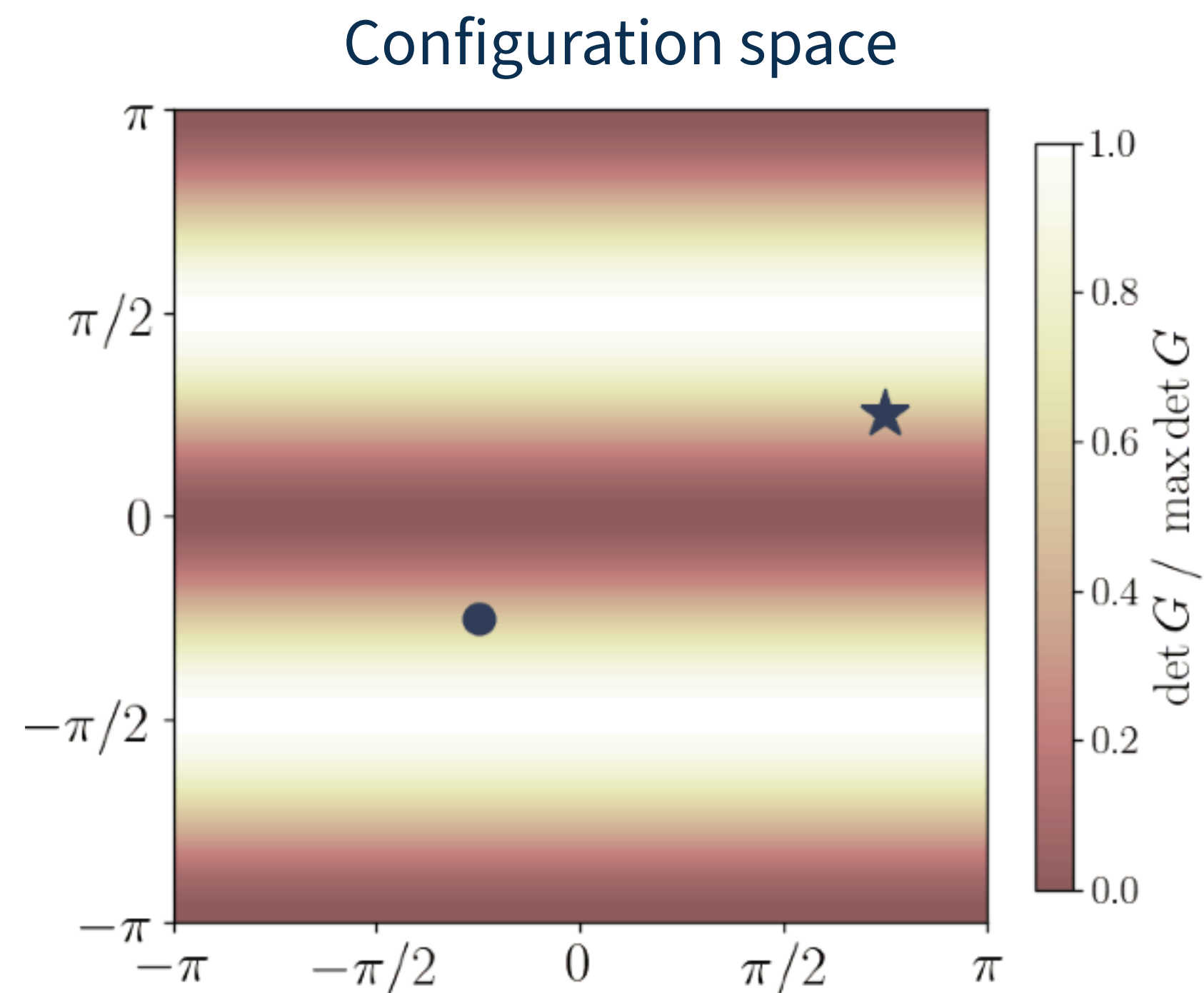






The simplest yet elegant anisotropic case

— **minimum-energy** motions, **kinetic-energy** Riemannian metric



RRT under the kinetic-energy metric*

$$T(q, \dot{q}) = \frac{1}{2} \dot{q}^\top \underbrace{M(q)}_{\text{inertia matrix}} \dot{q} \implies G(q) = M(q)$$

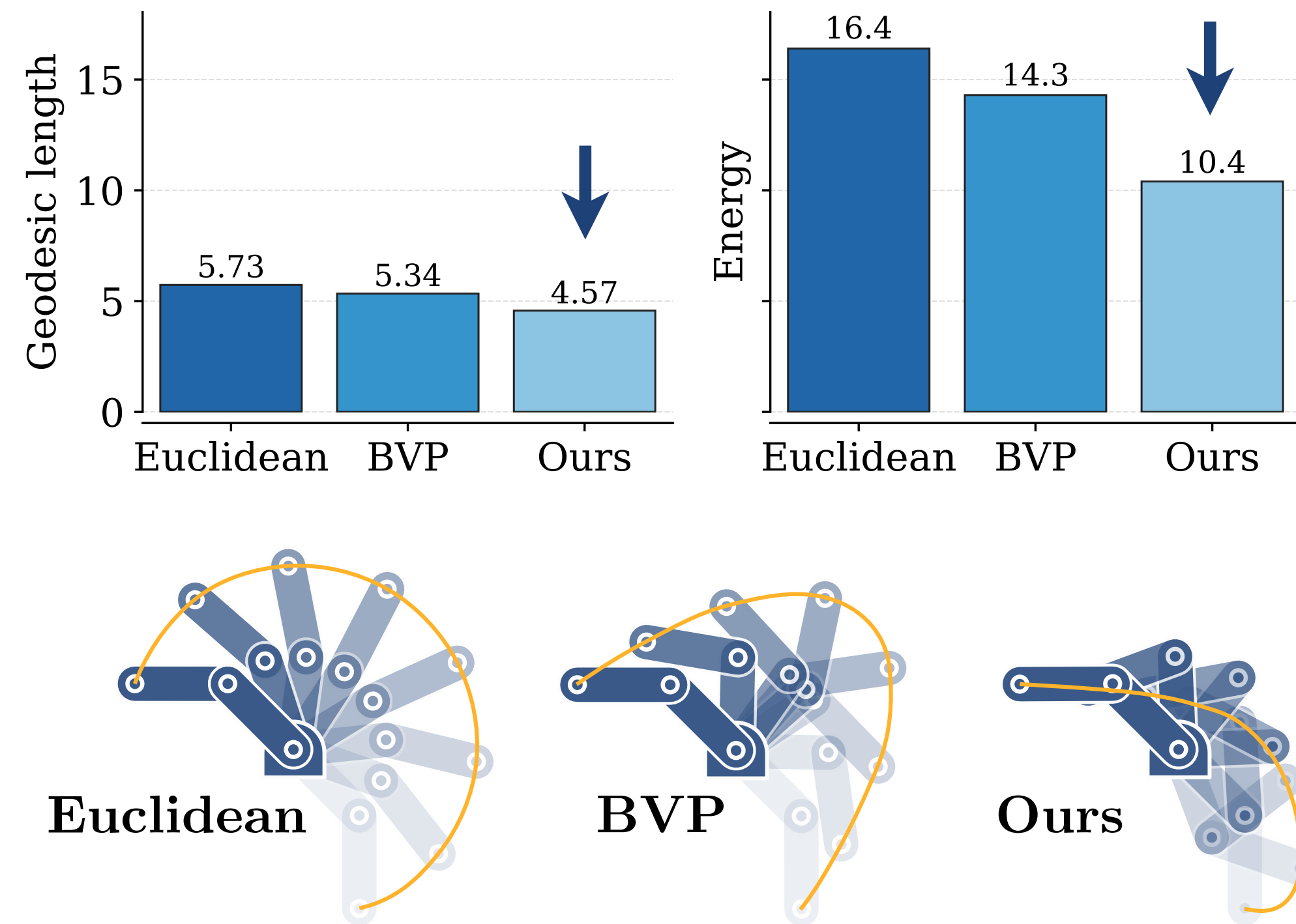
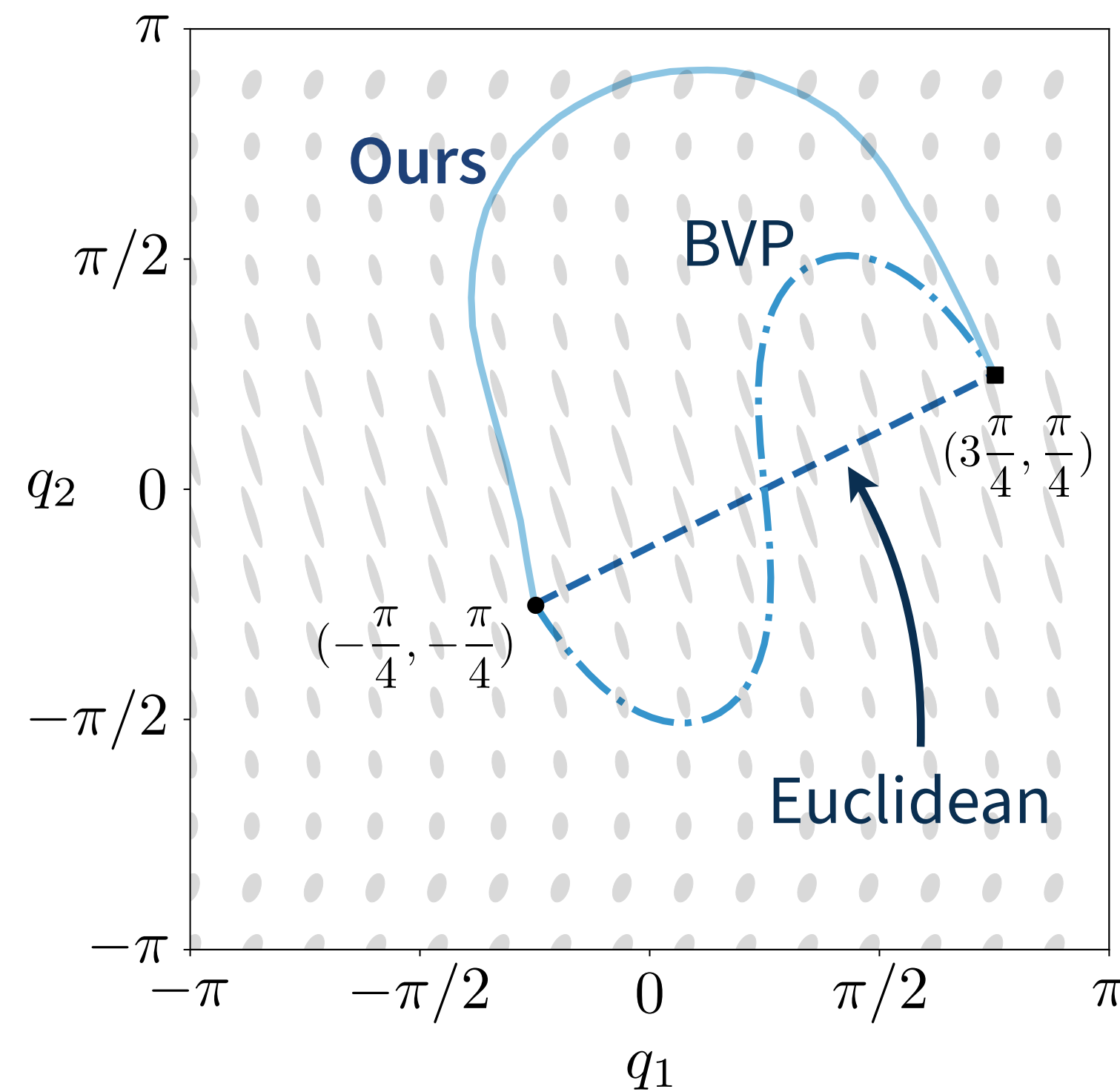
Task space



Euclidean vs. kinetic-energy

Shorter and lower-energy geodesics

— global search avoids the local-minimum trap



Numerical BVP solvers often need good initialization, else they tend to get stuck in local minima.